Course Chatbot

- CSV-based course chatbot (improved dataset)
- `train_model.py` to train the model
- Flask backend (`app.py`)
- HTML UI (`index.html`)
- Using `TfidfVectorizer` for better text handling

□ Project Structure

```
course-chatbot/
│
├── intents.csv
├── train_model.py
├── app.py
├── model.pkl
├── vectorizer.pkl
└── templates/
    └── index.html
```

---

1 □ `intents.csv` (Improved Dataset)

```
question,intent,response
What courses do you offer,course_list,"We offer Python, Data Science, Web
Development, MS Office, Tally with GST, Advanced Excel, Full Stack
Development, and Digital Marketing courses."
List all courses,course_list,"We offer Python, Data Science, Web Development,
MS Office, Tally with GST, Advanced Excel, Full Stack Development, and
Digital Marketing courses."
Which courses are available,course_list,"We offer Python, Data Science, Web
Development, MS Office, Tally with GST, Advanced Excel, Full Stack
Development, and Digital Marketing courses."

Tell me about Python course,python_info,"Python course covers basics to
advanced topics including projects."
Give details of Python course,python_info,"Python course covers basics to
advanced topics including projects."
What is included in Python course,python_info,"Python course covers basics to
advanced topics including projects."

What is fee for Python course,python_fee,"Python course fee is ₹10,000."
Python course fees,python_fee,"Python course fee is ₹10,000."
How much is Python course,python_fee,"Python course fee is ₹10,000."

Python course duration,python_duration,"Duration of Python course is 3
months."
How long is Python course,python_duration,"Duration of Python course is 3
```

months."
What is duration of Python course,python_duration,"Duration of Python course is 3 months."

Do you offer Data Science course,data_info,"Yes, we offer Data Science course with practical training."
Tell me about Data Science course,data_info,"Yes, we offer Data Science course with practical training."

Data Science course fees,data_fee,"Data Science course fee is ₹25,000."
What is fee for Data Science,data_fee,"Data Science course fee is ₹25,000."

Tell me about MS Office course,ms_info,"MS Office includes Word, Excel, PowerPoint and basic computer skills."
MS Office course details,ms_info,"MS Office includes Word, Excel, PowerPoint and basic computer skills."

MS Office fees,ms_fee,"MS Office course fee is ₹5,000."
What is fee for MS Office,ms_fee,"MS Office course fee is ₹5,000."

MS Office duration,ms_duration,"Duration is 1 month."
How long is MS Office course,ms_duration,"Duration is 1 month."

Tell me about Tally with GST,tally_info,"Tally course covers accounting, GST filing and business management."
Tally course details,tally_info,"Tally course covers accounting, GST filing and business management."

Tally course fees,tally_fee,"Tally course fee is ₹8,000."
What is fee for Tally,tally_fee,"Tally course fee is ₹8,000."

Tally course duration,tally_duration,"Duration is 2 months."
How long is Tally course,tally_duration,"Duration is 2 months."

Tell me about Advanced Excel,excel_info,"Advanced Excel covers formulas, pivot tables, dashboards and automation."
Advanced Excel details,excel_info,"Advanced Excel covers formulas, pivot tables, dashboards and automation."

Advanced Excel fees,excel_fee,"Advanced Excel course fee is ₹7,000."
What is Excel course fee,excel_fee,"Advanced Excel course fee is ₹7,000."

Excel duration,excel_duration,"Duration is 1.5 months."
How long is Excel course,excel_duration,"Duration is 1.5 months."

Tell me about Full Stack Development,fullstack_info,"Full Stack includes frontend and backend technologies like HTML, CSS, JavaScript and Python."
Full stack course details,fullstack_info,"Full Stack includes frontend and backend technologies like HTML, CSS, JavaScript and Python."

Full stack fees,fullstack_fee,"Full Stack course fee is ₹30,000."
What is fee for full stack,fullstack_fee,"Full Stack course fee is ₹30,000."

Full stack duration,fullstack_duration,"Duration is 6 months."
How long is full stack course,fullstack_duration,"Duration is 6 months."

Tell me about Digital Marketing,dm_info,"Digital Marketing includes SEO,

```
social media marketing and ads."
Digital marketing course details,dm_info,"Digital Marketing includes SEO,
social media marketing and ads."

Digital marketing fees,dm_fee,"Digital Marketing course fee is ₹20,000."
What is fee for digital marketing,dm_fee,"Digital Marketing course fee is
₹20,000."

Digital marketing duration,dm_duration,"Duration is 3 months."
How long is digital marketing course,dm_duration,"Duration is 3 months."

Hi,greeting,"Hello! How can I help you?"
Hello,greeting,"Hi there! Ask me anything about courses."
Hey,greeting,"Hello! How can I help you?"
```

---

2 `train_model.py` (Train Model)

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
import joblib

# Load CSV
data = pd.read_csv('intents.csv')

# Drop empty rows
data = data.dropna(subset=['question'])
data = data.reset_index(drop=True)

# Features & labels
X = data['question'].astype(str)
y = data['intent']

# Vectorizer
vectorizer = TfidfVectorizer()
X_vec = vectorizer.fit_transform(X)

# Model
model = LogisticRegression(max_iter=1000)
model.fit(X_vec, y)

# Save
joblib.dump(model, 'model.pkl')
joblib.dump(vectorizer, 'vectorizer.pkl')

print("Model trained successfully!")
```

Chatbot into a **ChatGPT-style web UI** with:

- A **scrollable chat window**
- **User and bot messages** clearly separated
- **Typing animation** for bot responses
- **Fallback response** if the bot doesn't understand

I'll give the **full updated HTML + minor Flask tweaks**.

---

1 `app.py` (Add fallback & JSON response)

```
from flask import Flask, render_template, request, jsonify
import joblib
import pandas as pd

app = Flask(__name__)

# Load model and vectorizer
model = joblib.load('model.pkl')
vectorizer = joblib.load('vectorizer.pkl')

# Load CSV responses
data = pd.read_csv('intents.csv')

@app.route('/')
def home():
    return render_template('chat.html')  # renamed template

@app.route('/get', methods=['POST'])
def chatbot():
    user_input = request.form['message']

    # Transform input
    input_vec = vectorizer.transform([user_input])

    # Predict intent
    intent = model.predict(input_vec)[0]

    # Get response
    responses = data[data['intent'] == intent]['response'].values

    if len(responses) > 0:
        response = responses[0]
    else:
        response = "Sorry, I didn't understand. Can you rephrase?"

    return jsonify({'response': response})
```

```
if __name__ == "__main__":
    app.run(debug=True)
```

---

2 New `templates/index.html` (ChatGPT-style UI)

```
<!DOCTYPE html>
<html>
<head>
    <title>Course Enquiry Chatbot</title>
    <style>
        body { font-family: Arial; background: #f0f0f0; }
        .chat-container { width: 400px; margin: 50px auto; background: #fff;
border-radius: 8px; padding: 20px; box-shadow: 0 0 10px #aaa; }
        .chat-box { height: 400px; overflow-y: auto; border: 1px solid #ddd;
padding: 10px; }
        .user-msg { text-align: right; margin: 10px 0; }
        .bot-msg { text-align: left; margin: 10px 0; }
        .msg { display: inline-block; padding: 8px 12px; border-radius: 15px;
max-width: 80%; }
        .user-msg .msg { background: #0084ff; color: white; }
        .bot-msg .msg { background: #e5e5ea; color: black; }
        #userInput { width: 80%; padding: 10px; }
        #sendBtn { width: 18%; padding: 10px; background: #0084ff; color:
white; border: none; cursor: pointer; }
    </style>
</head>
<body>

<div class="chat-container">
    <h3>Course Enquiry Bot</h3>
    <div class="chat-box" id="chatBox"></div>
    <input type="text" id="userInput" placeholder="Ask something...">
    <button id="sendBtn">Send</button>
</div>

<script>
const chatBox = document.getElementById('chatBox');
const input = document.getElementById('userInput');
const btn = document.getElementById('sendBtn');

function appendMessage(sender, message) {
    const div = document.createElement('div');
    div.classList.add(sender + '-msg');
    div.innerHTML = `<span class="msg">${message}</span>`;
    chatBox.appendChild(div);
    chatBox.scrollTop = chatBox.scrollHeight;
}

function sendMessage() {
    const message = input.value;
    if (message.trim() === '') return;

    appendMessage('user', message);
```

```
    input.value = '';

    // Typing animation
    appendMessage('bot', 'Typing...');
    const typingDiv = chatBox.lastChild;

    fetch("/get", {
        method: "POST",
        headers: { "Content-Type": "application/x-www-form-urlencoded" },
        body: "message=" + encodeURIComponent(message)
    })
    .then(response => response.json())
    .then(data => {
        typingDiv.remove(); // remove typing
        appendMessage('bot', data.response);
    });
}

btn.addEventListener('click', sendMessage);
input.addEventListener('keypress', function(e){
    if(e.key === 'Enter') sendMessage();
});
</script>

</body>
</html>
```

---

3 Features Added

- Scrollable chat window
- User and bot messages in bubbles
- “Typing...” animation before bot responds
- Enter key also sends message
- Fallback message if intent not found

---

4 Run Steps

1. Make sure your **model is trained**:

```
python train_model.py
```

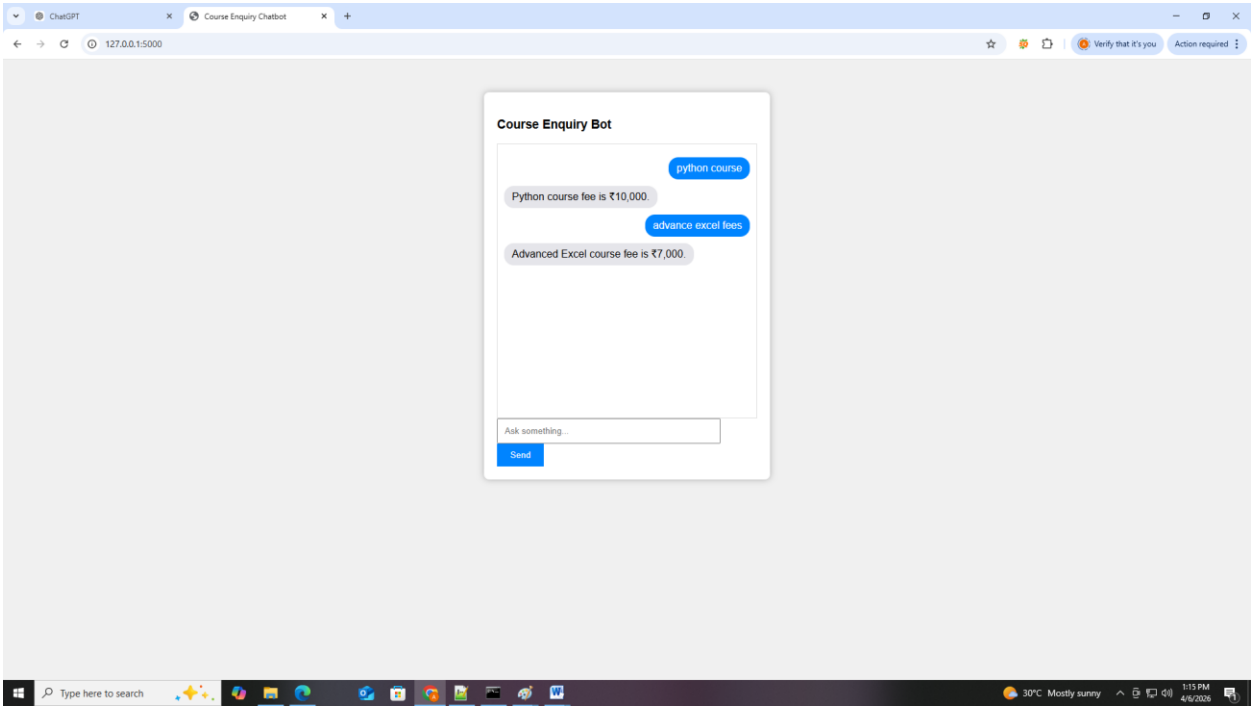
2. Run Flask:

```
python app.py
```

3. Open browser:

http://127.0.0.1:5000/

Output:-



Locally:-

```
pip install -r requirements.txt
```

1. Push project on github using git:

```
git init
git add .
git commit -m "initial commit"
```
git push -u origin main

Deploy to Render

1. Sign up at render.com
2. Connect your GitHub repo
3. Set start command: `gunicorn app:app`
4. Deploy → Get public URL

Let's focus on **why `TfidfVectorizer` is used in your chatbot project** and how it works step by step.

---

1 What is `TfidfVectorizer`?

`TfidfVectorizer` is a **text feature extractor** from scikit-learn. It converts **text (words)** into **numbers** so that machine learning models can understand them.

- TF = **Term Frequency** → How often a word appears in a sentence.
- IDF = **Inverse Document Frequency** → Words that appear in **many sentences** (like "is", "the") get lower weight.
- TF-IDF = TF × IDF → Emphasizes **important words** for each sentence.

2 Why we use it in your chatbot

Your chatbot gets **user input in text form**, like:

```
"What is the fee for Python course?"
```

But your **ML model cannot understand plain text**. It only works with **numbers**.

`TfidfVectorizer` solves this by:

1. Learning a **vocabulary** from all your training questions.
2. Converting each new user input into a **vector of numbers** representing word importance.

Example

```
from sklearn.feature_extraction.text import TfidfVectorizer

# Training questions
questions = [
    "What courses do you offer?",
    "Tell me about Python course",
    "What is the fee for Python course?"
]

# Initialize vectorizer
vectorizer = TfidfVectorizer()

# Learn vocabulary & transform training data
X = vectorizer.fit_transform(questions)

print("Vocabulary:", vectorizer.get_feature_names_out())
print("Vector shape:", X.shape)
```

Output:

```
Vocabulary: ['about', 'courses', 'do', 'fee', 'offer', 'python', 'tell',
'the', 'what']
Vector shape: (3, 9)
```

- Each sentence is now a **9-dimensional vector**.
- The ML model can now **classify which intent** (like `course_list` or `python_fee`) the question belongs to.

---

3 How it fits in your chatbot pipeline

1. **Training phase:**

```
vectorizer = TfidfVectorizer()
X_train_vec = vectorizer.fit_transform(X_train_texts)  # convert all
questions to numbers
model.fit(X_train_vec, y_train)  # train classifier
```

2. **Save vectorizer for deployment:**

```
joblib.dump(vectorizer, 'vectorizer.pkl')
```

3. **Prediction phase (Flask app):**

```
vectorizer = joblib.load('vectorizer.pkl')
model = joblib.load('model.pkl')

user_input = "Tell me about Python course"
input_vec = vectorizer.transform([user_input])  # convert user input to
vector
prediction = model.predict(input_vec)[0]        # predict intent
```

 Without `TfidfVectorizer`, the model **cannot convert text to numbers**, and it will **fail to understand any user input**.